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1. Database is an organized collection of data stored electronically for easy access and management. It is essential because it allows fast storage, retrieval and updating of information. Examples: Banking systems, online shopping websites.

2. Data warehouse stores large amounts of historical data for analysis and reporting. Differences, are databases handle daily operations; data warehouses handle analysis. It is used to support business decisions and trends analysis.

3. Data lake stores raw data in its original format (structured and unstructured). The difference is that data warehouses store processed data, data lakes stores raw data. The use is when a ~~ft~~ business needs flexibility and handles big data. for example, AI, analytics.)

4. Data mart is a smaller section of a data warehouse focused on a specific department. It comes from a data warehouse. It is used by departments like sales, finance or HR.

5. Structured data - Organized in tables (eg Excel, spreadsheet)
Semi-Structured - Some structure (eg JSON files)
Unstructured data - No fixed format (eg

6. Data modeling is designing how data is structured and related. ~~In~~ It ensures efficient storage and easy ~~an~~ access. Types include Conceptual model (high-level design) and Logical model

13.) A subset of machine learning that uses complex models called neural networks. It is part of ml which is part of AI. Neural networks are systems inspired by the human brain; they process data through layers to recognize patterns.

14. Natural language processing allows machines to understand and respond to human language. Examples include voice assistants like google assistant. Predictive text on smartphones. Translation apps like google translate.

15. An advanced AI model trained on large on large text data to understand and generate language. LLM handle complex language tasks; standard ML models are usually task-specific. Examples include ChatGpt answers questions. GPt-4 advanced text generation.

16. AI that creates new content (text, images etc). How it works, it learns patterns from data and generates similar new outputs. Tools include ChatGpt, Dall-E and Midjourny.

17. A collection of data used to train an AI model. High-quality data leads to accurate results. Biased or incorrect data can cause unfair or wrong predications.

18. AI that enables machines to interpret images and videos. Uses algorithms to detect patterns like shapes and objects. Examples facial recognition systems. Self-driving cars detecting roads and obstacles.

19. An API is a tool that allows different software systems to

(detailed structure).

7. Data mining is analyzing data ~~is~~ to find patterns and trends. It Helps with decision-making in business. Examples, customer purchase recommendations, fraud detection.

8. DBMS is software used to create and manage databases. MySQL - Web applications. PostgreSQL - Advanced data analysis. Oracle - Large enterprise systems. MongoDB - Flexible data (NoSQL, unstructured data)

9. Extract - Get data from sources. Transform - Clean and format data. Load - Store data in a system. Importance - Ensures data is accurate and ready for analysis.

10. CSV - Simple text tables (small datasets). JSON - Flexible, semi structured data (APIs). Parquet - Efficient for big data (analytics). Avro - Good for data streaming and storage.

11. AI is the ability of machines to perform tasks that normally require human intelligence. Narrow AI designed for one ~~task~~ task. General AI can perform any intellectual task like a human.

12. Machine learning is a type of AI where systems learn from data instead of being explicity programmed. Traditional programming uses fixed rules, machine learns patterns from data. Supervised: Learns from labeled data (eg spam detection). Unsupervised: Finds pattern in unlabeled data (eg customer grouping). Reinforcement: Learns through rewards and penalties (eg game-playing AI).

Docs).

26. Tracks changes in code over time. Git - popular system for managing code versions. Commit - A saved change. Branch - A separate version for development.

27. An interactive tool for writing and running code. Language - mostly Python. Combines code, results and notes in one place, great for data analysis and visualization.

28. Python is a high-level programming language known for simplicity and readability. It is easy to learn, large community, strong libraries for data and AI. Libraries include NumPy - numerical computing. Pandas - data analysis and tables. Matplotlib - data visualization. Tensorflow - AI and ~~deep~~ deep learning.

29. Protecting personal information from misuse. Prevents identity theft and protects rights. GDPR, ~~a~~ is a law regulating data protection in the European Union; applies to EU countries and any company handling EU citizens' data.

30. Guidelines for using data responsibly. Examples include selling user data without consent → violates privacy. Biased hiring algorithms → unfair decisions. Misusing facial recognition → invades personal rights.

31. Management of data quality, security and usage. Key components include data quality (accuracy), security (protection) policies and standards and compliance (legal rules).

32. When AI produces unfair or inaccurate results due to biased data. Example, facial recognition performing poorly on certain skin tones → can lead to wrongful identification.

33.

Role	Task	Skills	Tools
Data analyst	Analyze data & create reports.	Excel, SQL, visualization	Power BI, Tableau.
Data Scientist	Build predictive models	Statistics, ML, Coding.	Python, Scikit-learn
Data Engineer	Build data Pipelines	Databases, big data	SQL, Apache Spark
AI Engineer	Develop AI Systems	ML, deep learning	Python, Py Torch

34. BI is the use of tools and systems to analyze data and support decision-making. Tools include, microsoft Power BI which creates dashboards and reports. Tableu which visualizes data trends. They turn raw data into insights so business can make better decisions.

35. The most interesting concept was AI, especially how it can learn and make decisions. It is surprising how tools like ChatGpt can understand and generate human-like responses.

36. The most difficult concept was Deep Learning because of neural networks, I used YouTube tutorials and online articles to better understand it.

37. Data and AI will impact my future career by improving

communicate. Analogy, like a waiter taking your order to the kitchen and bringing back food. Enables apps to share data and services easily.

20. A unique code used to identify and authorize access to an API. Controls and traces usage. Store privately (e.g environment variables). If exposed, others can misuse the API or cause charges.

21. REST - Representational State Transfer (a way to design APIs) Get - Retrieve data. Post - Send/create data. Put - update data. Delete - Remove data.

22. JSON (JavaScript Object Notation). A lightweight format for storing and exchanging data. Easy to read and works across systems. Example "Name": "Ayanda" ; "Age": 21 ; "City": "Cape Town

23. Designing clear and effective prompts for AI tools. Better prompts equals better results. Examples: Poor - "write something about business". Better - "Write a 100-word summary on the importance of customer service in small businesses. Poor

24. Applies a function to each item in a list and returns a new list. Example : numbers = [1, 2, 3]

result = list(map(lambda x: x * 2, numbers))

Output: [2, 4, 6]

25. Delivering computing services over the internet. IaaS: infrastructure (e.g. Amazon Web Services). PaaS: Platform for development (e.g Google app engine). SaaS: Ready-to-use software (e.g Google

decision-making and efficiency. for example, AI tools can automate tasks, while data analysis can help businesses understand customers better.

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